CASE-BASED SCENARIO
WINE QUALITY PREDICTION

Using Ridge, Lasso and Elastic Net Regression

1. Case Background

A wine producer receives different batches of red wine that need to be assessed for quality. Traditional wine-quality assessment relies on sensory evaluation by trained tasters, which can be subjective and time-consuming. The producer has laboratory measurements describing the physicochemical properties of each wine sample and wants to use these measurements to support quality assessment.

2. Real-World Problem

The main problem is to predict the quality score of a red wine sample from its physicochemical properties. Wine quality is influenced by multiple measurable chemical characteristics, and the relationship between these properties and sensory quality can be complex. Therefore, the problem is formulated as a supervised regression problem, where the numerical quality score is predicted from measurable physicochemical features.

3. Available Data

The project uses the WineQT.csv dataset.

Dataset Property	Details
Records	1,143
Columns	13
Predictor Variables	11 physicochemical features
Target Variable	quality
Identifier	Id
Quality Range	3–8

Input features: Fixed acidity, volatile acidity, citric acid, residual sugar, chlorides, free sulfur dioxide, total sulfur dioxide, density, pH, sulphates and alcohol. The Id column is excluded from model training because it is only an identifier.

4. Proposed ML Solution

Three regularized regression algorithms are used to predict the numerical wine-quality score:

Ridge Regression: Uses L2 regularization to shrink coefficient magnitudes and help control multicollinearity.

Lasso Regression: Uses L1 regularization and can shrink some coefficients to zero, providing feature-selection behavior.

Elastic Net Regression: Combines L1 and L2 regularization, providing a balance between feature selection and coefficient stability.

The data is divided into training and testing sets using an 80/20 split with random_state = 42. StandardScaler is used for feature scaling, and the models are tuned using 5-fold cross-validated GridSearchCV. Performance is evaluated using MAE, MSE, RMSE and R^2.

5. Expected Output

For a given wine sample's physicochemical measurements, the system produces a predicted wine-quality score. The three regression models can then be compared based on their prediction performance.

Physicochemical Measurements → Regression Model → Predicted Quality Score

6. Case Study Flow

Wine Quality Dataset

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Data Preprocessing

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Exploratory Data Analysis (EDA)

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Feature / Target Separation

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Remove Id

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80/20 Train-Test Split

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Standard Scaling

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Ridge | Lasso | Elastic Net

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Prediction

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Model Evaluation

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MAE | MSE | RMSE | R^2

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Model Comparison

7. Project Implementation

The project includes practical EDA implementation and model implementation for Ridge, Lasso and Elastic Net Regression. The modeling stage includes preprocessing, hyperparameter tuning, prediction and evaluation using the specified regression metrics.

8. Case Study Objective

The objective is to demonstrate how measurable physicochemical properties of red wine can be used in a machine-learning regression workflow to estimate wine quality and compare different regularization strategies.